Financial AI Analyzer – Transforming Complex Data into Actionable Insights

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1. INTRODUCTION

1.1 Project Overview

Financial AI Analyzer is an intelligent financial analysis application designed to analyze financial data and generate structured insights for users. The system is developed using Python with an interactive interface built using Streamlit and integrates the powerful Gemini Flash AI model to interpret financial information and generate meaningful analytical outputs.

The primary objective of the system is to assist users in understanding financial data by automatically analyzing uploaded financial information and producing structured insights such as spending patterns, income distribution, financial trends, savings potential, and risk indicators. The system leverages the reasoning capabilities of Gemini Flash to process financial inputs and generate contextual explanations and recommendations.

Unlike traditional financial tools that require manual analysis or complex spreadsheets, Financial AI Analyzer uses generative AI to interpret financial data and present simplified insights that help users make better financial decisions. The system can evaluate financial records and provide useful information such as expense breakdowns, savings opportunities, budget planning suggestions, and investment awareness guidance.

The application is divided into two major modules:

Primary Module – Financial Data Analysis

This module allows users to upload financial data or enter financial details such as income, expenses, and spending categories. The AI system analyzes the data and generates structured financial insights including spending summaries, financial health indicators, and budget recommendations.

Enhancement Module – Financial Advisory Assistant

This module enables users to interact with the system through text-based queries related to personal finance. Users can ask questions about budgeting, saving strategies, expense

management, or financial planning. The AI assistant processes these queries and provides informative financial guidance and suggestions.

The integration of AI reasoning with structured prompt design ensures that the responses remain consistent, understandable, and useful for practical financial decision-making.

1.2 Purpose

The purpose of the Financial AI Analyzer project is to design and implement an intelligent AI-based financial analysis system that helps users better understand and manage their financial activities.

The system aims to:

- Analyze financial data and generate structured insights
- Identify spending patterns and financial trends
- Provide budgeting and savings recommendations
- Assist users in understanding their financial health
- Support financial planning through intelligent analysis
- Provide an interactive financial advisory assistant for user queries

By combining artificial intelligence with financial analysis techniques, the Financial AI Analyzer demonstrates how AI-powered systems can simplify complex financial data and assist users in making informed financial decisions.

The project highlights the practical application of modern AI technologies such as Gemini Flash in financial intelligence systems and decision-support tools.

2. IDEATION PHASE

2.1 Problem Statement

Managing personal and organizational finances can be complex due to the large amount of financial data generated through daily transactions, expenses, and income sources. Many individuals and small businesses struggle to effectively analyze their financial data and understand their financial health.

Users often face difficulties in:

- Understanding their spending patterns.
- Identifying unnecessary expenses.

- Tracking income and expenditure effectively.
- Creating realistic budgets.
- Making informed financial decisions.
- Recognizing potential savings opportunities.

Existing financial management tools often require manual analysis using spreadsheets or provide limited insights. Users must interpret raw data themselves, which can be time-consuming and confusing, especially for individuals without financial expertise.

There is a need for an intelligent system that can:

- Accept financial data as input.
- Automatically analyze financial transactions and spending behavior.
- Generate structured financial insights.
- Provide budgeting and savings recommendations.
- Offer interactive financial advisory support.

The Financial AI Analyzer addresses this challenge by integrating AI-based reasoning using Gemini Flash to analyze financial data and generate structured financial insights that help users better understand and manage their finances.

2.2 Empathy Map Canvas

Target Users

- Individuals managing personal finances.
- Students learning financial planning.
- Working professionals tracking monthly expenses.
- Small business owners analyzing financial records.
- Users seeking financial advice and budgeting support.

What Users Think

- “Where is most of my money being spent?”
- “Am I saving enough every month?”
- “How can I reduce unnecessary expenses?”
- “What is the best way to manage my budget?”

What Users Feel

- Confused about their financial habits.
- Overwhelmed by financial records and transactions.
- Concerned about long-term savings and financial stability.

What Users Do

- Track expenses manually.
- Use spreadsheets for budgeting.
- Search online for financial advice.
- Use multiple apps to manage financial information.

Pain Points

- Lack of automated financial insights.
- Difficulty analyzing large amounts of financial data.
- Limited understanding of spending patterns.
- Time-consuming manual financial analysis.

The Financial AI Analyzer simplifies this process by automatically analyzing financial data and generating clear, structured insights using AI-driven analysis.

2.3 Brainstorming

During the ideation phase, several technical approaches were evaluated to determine the best solution for building an intelligent financial analysis system.

Approach 1: Static Financial Analysis System

This approach involves storing financial data in a database and generating reports using predefined rules.

Limitation:
Requires manual rule creation and cannot generate contextual explanations.

Approach 2: Traditional Data Analysis Tools

Financial data could be analyzed using statistical models or traditional analytics tools.

Limitation:
These tools provide numerical outputs but lack conversational insights and intelligent recommendations.

Approach 3: AI-Based Financial Analysis

Using a generative AI model allows the system to:

- Interpret financial data.
- Generate contextual explanations.

- Provide budgeting and saving recommendations.
- Answer user financial queries interactively.
- Produce structured and understandable insights.

Final Decision

After evaluating different approaches, the project team decided to implement an AI-powered financial analysis system using **Gemini Flash** integrated with **Python** and **Streamlit**.

This approach enables intelligent financial data interpretation, automated insights, and interactive financial advisory features.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. The user opens the Financial AI Analyzer application.
2. The user uploads financial data such as transaction records or financial statements.
3. The system processes the data using the integrated AI analysis module.
4. The AI model analyzes spending patterns, trends, and anomalies.
5. The system generates structured financial insights and visualizations.
6. The user reviews reports and uses the insights for better financial decision-making.

3.2 Solution Requirements

Functional Requirements

- Upload financial datasets (CSV/Excel).
- Data preprocessing and cleaning.
- AI-based financial analysis.
- Dashboard visualization of insights
- Report generation.
- User interaction with analytics tools.
- Secure data handling.

Non-Functional Requirements

- Response time below 5 seconds for analysis tasks.

- Secure storage and processing of financial data.
- User-friendly interface for data visualization.
- High system reliability and scalability.

3.3 Data Flow Diagram

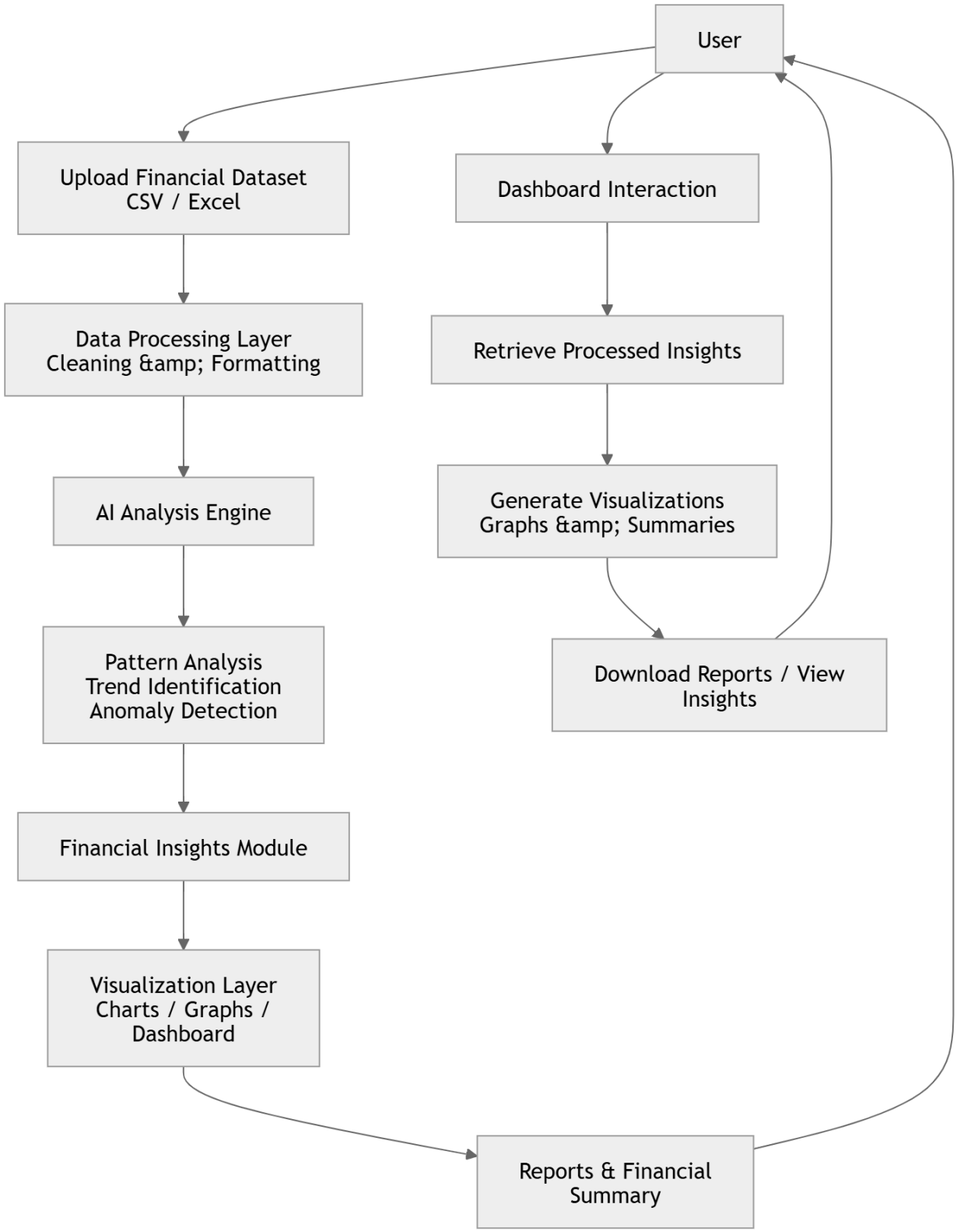
The Financial AI Analyzer follows a data processing architecture that includes data ingestion, AI-based analysis, and visualization.

Primary Flow – Financial Data Analysis

1. The user uploads financial data through the application interface.
2. The data is processed in the Data Processing Layer, where cleaning and formatting are performed.
3. The processed data is sent to the AI Analysis Engine.
4. The AI engine analyzes spending patterns, anomalies, and financial trends.
5. The results are organized in the Financial Insights Module.
6. The insights are displayed in charts, dashboards, and reports.

Secondary Flow – User Interaction

1. The user interacts with the dashboard to explore financial insights.
2. The system retrieves processed data from the analysis module.
3. Visualizations such as graphs and summaries are generated.
4. The user can download reports or review insights for decision-making.


User
Upload Financial Dataset
CSV / Excel
Dashboard Interaction
Data Processing Layer
Cleaning & Formatting
Retrieve Processed Insights
AI Analysis Engine
Generate Visualizations
Graphs & Summaries
Pattern Analysis
Trend Identification
Anomaly Detection
Download Reports / View
Insights
Financial Insights Module
Visualization Layer
Charts / Graphs /
Dashboard
Reports & Financial
Summary

3.4 Technology Stack

- **Programming Language: Python**
  - Python is used as the core programming language for implementing data processing, AI analysis, and backend functionality. Its extensive ecosystem supports data analysis, machine learning, and visualization.
- **Frontend Framework: Streamlit**
  - Streamlit is used to build the interactive web interface of the Financial AI Analyzer. It enables users to upload financial datasets, view visualizations, and interact with analytical results.
- **AI Analysis Engine**
  - The AI module processes financial data to identify spending patterns, anomalies, and predictive insights, helping users understand their financial behavior.
- **Data Processing Libraries**
  - Libraries such as Pandas and NumPy are used for efficient data manipulation, cleaning, and statistical analysis.
- **Environment Management: dotenv**
  - The dotenv library is used to securely manage environment variables such as API keys and configuration parameters.
- **Deployment**
  - The system can run locally during development and can be deployed on cloud platforms for broader accessibility.

4. PROJECT DESIGN

4.1 Problem–Solution Fit

The Financial AI Analyzer system directly addresses the common challenges faced by individuals and organizations when analyzing financial data.

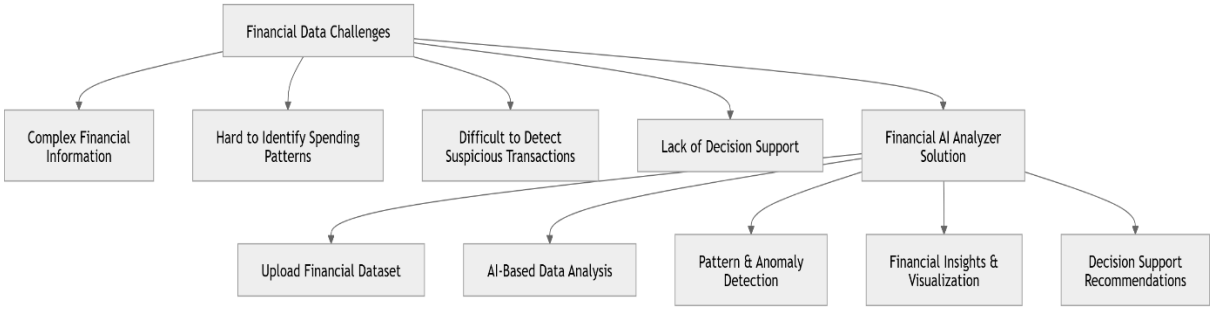
Many users struggle with:

- Understanding complex financial data.
- Identifying spending patterns and trends.
- Detecting unusual or suspicious transactions.
- Making informed financial decisions.

Traditional financial tools often provide only raw data without intelligent insights, which makes financial interpretation difficult for non-expert users.

The Financial AI Analyzer solves these problems by:

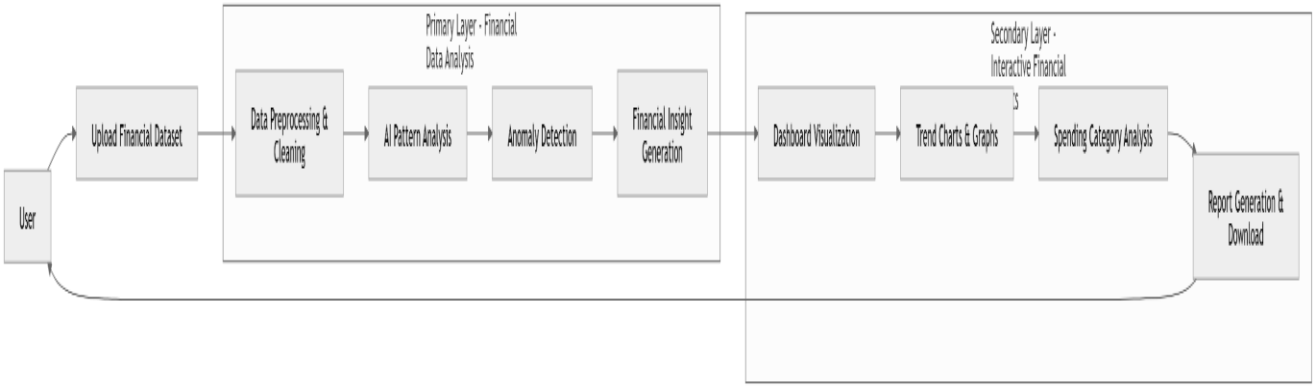
- Accepting financial datasets as input.
- Performing automated AI-driven analysis.
- Generating structured financial insights and visualizations.
- Providing decision-support recommendations.



The system uses structured data processing and AI analysis techniques to ensure accurate and consistent financial insights.

4.2 Proposed System

The Financial AI Analyzer system is divided into two main functional layers.



Primary Layer – Financial Data Analysis

This layer performs the core analytical operations.

Functions include:

- Uploading financial datasets (CSV/Excel).
- Data preprocessing and cleaning.

- AI-based financial pattern analysis.
- Detection of anomalies and irregular transactions.
- Generation of structured financial insights and reports.

Secondary Layer – Interactive Financial Insights

This layer enhances user interaction with the system.

Functions include:

- Allowing users to explore financial insights through dashboards.
- Providing graphical visualization of financial trends.
- Enabling users to analyze spending categories.
- Supporting report generation and download options.

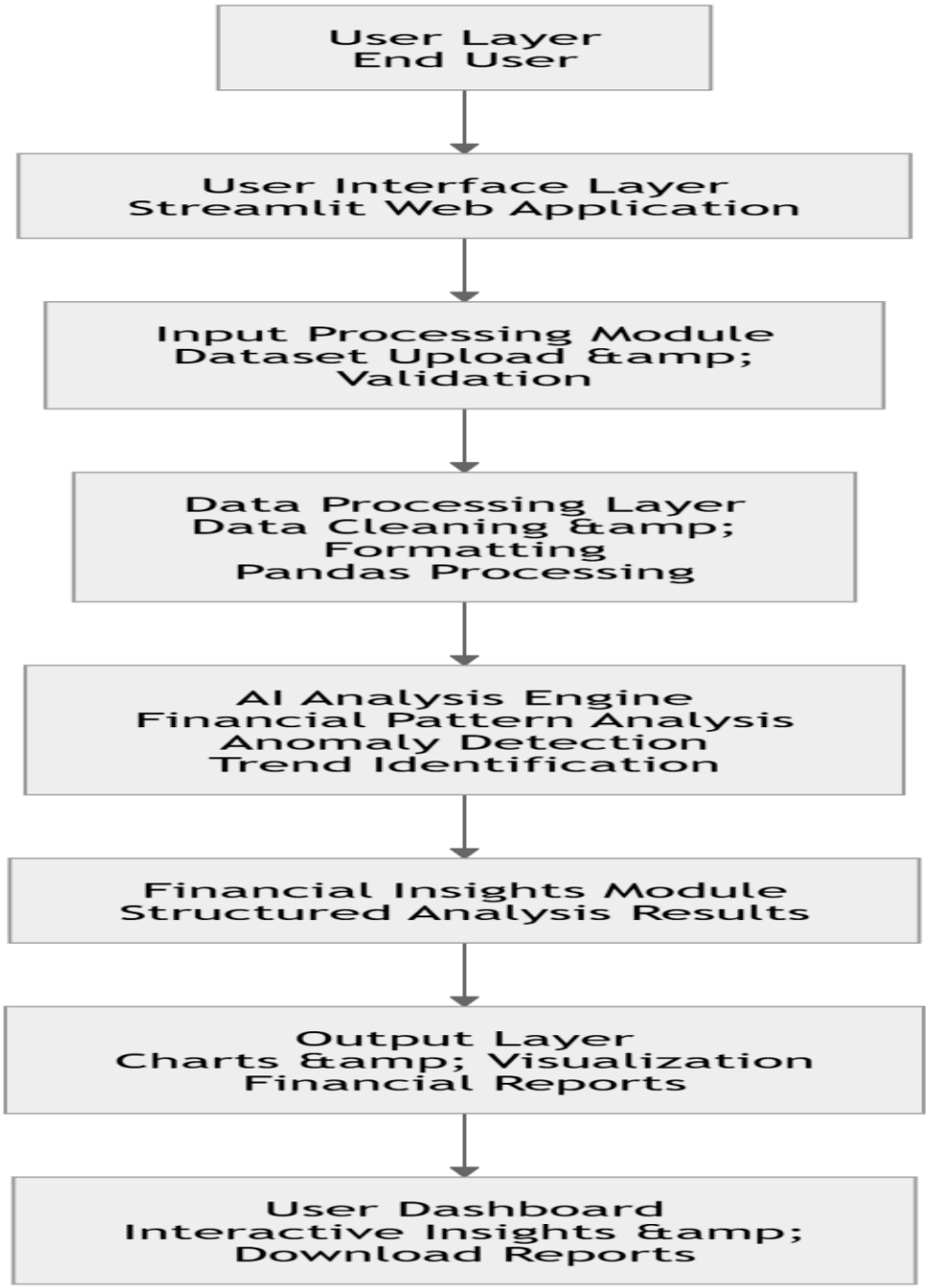
Together, these layers create an intelligent financial analysis environment that assists users in understanding their financial behavior and making informed decisions.

4.3 Solution Architecture

The Financial AI Analyzer system follows a simplified client–AI analytics architecture that enables efficient interaction between the user interface and the AI analysis engine.

The architecture consists of four major components:

1. User Layer
2. User Interface (UI) Layer
3. Processing Layer (AI Analysis Engine)
4. Output Layer



Architecture Flow Explanation

Step 1: User Interaction

The process begins when the user interacts with the system through the application interface.

The user can:

- Upload financial datasets (CSV/Excel files).

- Request analysis of financial records.
- View generated reports and insights.

This interaction is handled through the Streamlit-based user interface.

Step 2: Input Processing

Once the user uploads financial data, the system processes the input in the data processing module.

The system performs:

- Data cleaning and formatting.
- Conversion of uploaded files into structured data formats.
- Preparation of datasets for analysis.

The processed data is then forwarded to the AI Analysis Engine.

Step 3: AI Processing

The AI engine acts as the core intelligence module of the system.

It performs:

- Financial pattern analysis
- Transaction trend identification
- Detection of anomalies or unusual transactions
- Generation of meaningful financial insights

Using machine learning and data analysis techniques, the system interprets financial data and produces structured analytical outputs.

Step 4: Output Generation

The generated insights are returned to the user interface.

The UI performs the following tasks:

- Formats the analysis results.
- Displays charts and financial summaries.
- Presents insights in an easy-to-understand format.

- Allows users to download reports.

The final insights are then displayed to the user for financial decision-making.

5. PROJECT PLANNING AND SCHEDULING

Duration	Report Section	Activities Performed	Deliverables
Day 1 – Day 2	Introduction & Ideation Phase	Defined project overview and objectives. Identified challenges in financial data analysis. Prepared problem statement. Conducted brainstorming to finalize the concept of AI-based financial analysis.	Clear project concept and finalized scope.
Day 3 – Day 4	Requirement Analysis	Designed customer journey map. Identified functional and non-functional requirements. Created Data Flow Diagram (DFD). Finalized technology stack including Python, Streamlit, AI analysis libraries, and dotenv.	Finalized system requirements and workflow structure.
Day 5 – Day 6	Project Design	Established problem–solution fit. Designed financial data analysis module. Designed visualization and reporting module. Developed high-level system architecture.	Complete system design blueprint.
Day 7 – Day 8	Environment Setup & Integration	Configured development environment. Set up required libraries such as Pandas, NumPy, and Streamlit. Configured environment variables using dotenv. Performed initial testing of data processing modules.	Working system environment setup.
Day 9 – Day 11	Core Development (Analysis Module)	Implemented financial data upload interface. Developed data preprocessing pipeline. Integrated AI-based analysis logic. Generated structured financial insights.	Functional financial data analysis module.
Day 12 – Day 14	Visualization Module	Implemented charts and dashboards for financial insights. Added data visualization	Interactive financial dashboard.

Duration	Report Section	Activities Performed	Deliverables
		features for trends and spending patterns. Enabled report display in the interface.	
Day 15 – Day 16	Testing & Optimization	Tested system with multiple financial datasets. Verified accuracy of insights and anomaly detection. Optimized visualization and UI layout.	Stable and optimized application.
Day 17	Results & Documentation	Captured output screenshots. Finalized architecture diagrams. Prepared project documentation and report.	Submission-ready project.

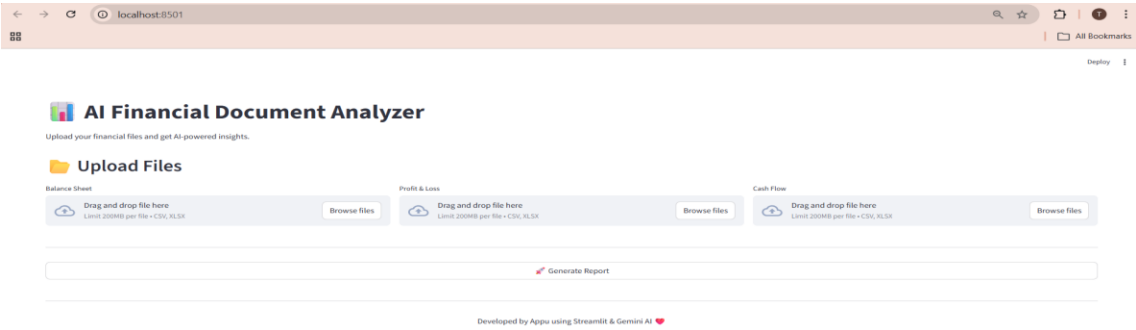
6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Testing

Functional testing ensures that each component of the Financial AI Analyzer system works according to the defined requirements.

6.1.1 Dataset Upload Validation

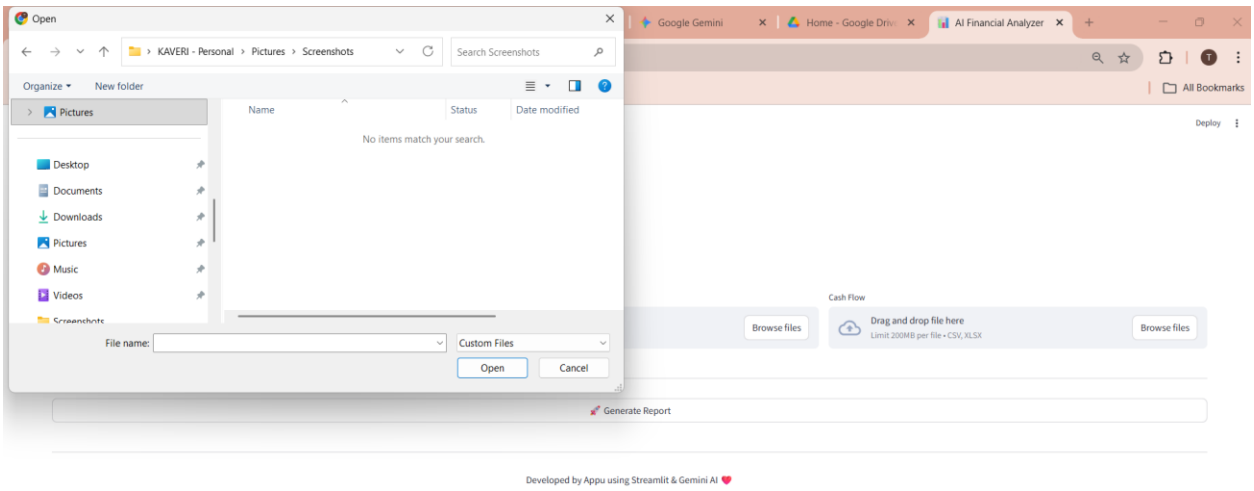
- **Test Case 1:** Analyze Button Visibility Without Dataset Upload
  - **Input:** No dataset uploaded
  - **Expected Result:** The "Analyze Financial Data" button should not be visible until a dataset is uploaded
  - **Actual Result:** The analyze button appears only after uploading a dataset and is not visible initially
  - **Status:**Passed



Before uploading the dataset, the Analyze Financial Data button is hidden. After uploading the dataset, the button becomes visible.

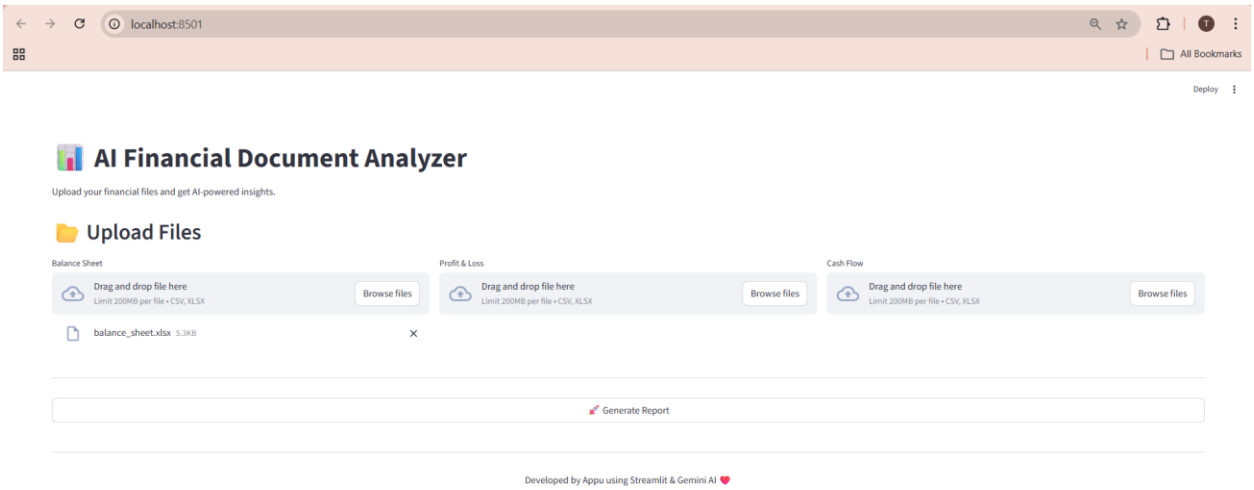
6.1.2 File Format Validation

- **Test Case 2:** Invalid File Format Upload
  - **Input:** Upload a non-supported file (e.g., .pptx file)
  - **Expected Result:** System should reject the file and display an error message indicating unsupported file type
  - **Actual Result:** Application displays an error message stating that the file format is not supported
  - **Status:** Passed



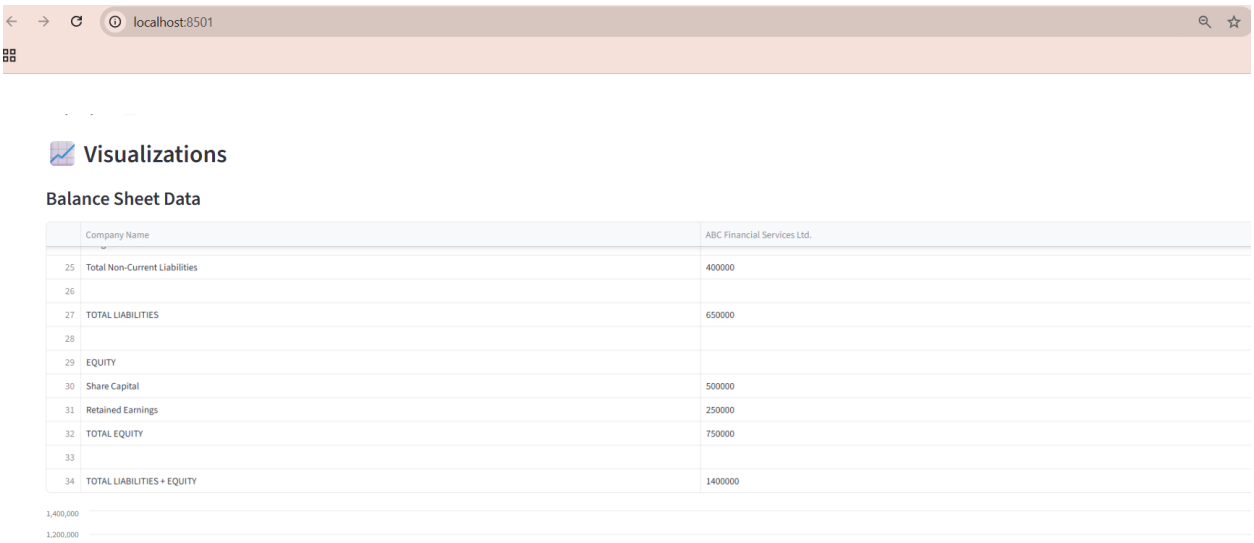
6.1.3 Dataset Upload Functionality

- **Test Case 3:** Valid Financial Dataset Upload
  - **Input:** Upload a CSV or Excel financial dataset
  - **Expected Result:** Dataset should load successfully and preview should be displayed
  - **Actual Result:** Dataset loaded successfully and preview table displayed
  - **Status:** Passed



6.1.4 Financial Data Analysis Test

- **Test Case 4: Financial Data Analysis Execution**
  - **Input:** Upload financial transaction dataset
  - **Expected Result:**
    1. Spending categories identified
    2. Financial trends detected
    3. Summary statistics generated
  - **Actual Result:** System successfully generated structured financial insights
  - **Status:** Passed



6.1.5 Visualization Module Testing

- **Test Case 5: Financial Chart Generation**
  - **Input:** Uploaded financial dataset
  - **Expected Result:**
    1. Charts representing spending distribution
    2. Graphs showing financial trends
  - **Actual Result:** Visual charts successfully displayed on the dashboard
  - **Status:** Passed

Profit & Loss Data

	Year	Revenue	Expense	Profit
0	2022	50000	30000	20000
1	2023	70000	40000	30000
2	2024	90000	55000	35000

6.1.6 Data Summary Generation

- **Test Case 7: Financial Summary Report**
  - **Input:** Financial dataset
  - **Expected Result:**
    1. Total income calculated
    2. Total expenditure calculated
    3. Balance summary generated
  - **Actual Result:** Structured financial summary report generated successfully
  - **Status:** Passed

A	B	C	D
Year	Revenue	Expense	Profit
2022	50000	30000	20000
2023	70000	40000	30000
2024	90000	55000	35000

6.2 Performance Testing

6.2.1 Response Time Analysis

• Average dataset analysis response time: **3–6 seconds**

The response time includes:

- Dataset upload and parsing
- Data preprocessing
- AI analysis execution
- Visualization rendering

The system maintained stable performance across multiple datasets.

6.2.2 System Stability Testing

- Multiple datasets were uploaded consecutively.
- The system processed repeated analysis requests without crashing.
- The application remained stable during extended use.

6.2.3 System Reliability

- No interface freezing occurred during analysis.
- Loading indicators were displayed during processing.
- Proper error messages appeared for invalid inputs.

7. RESULTS

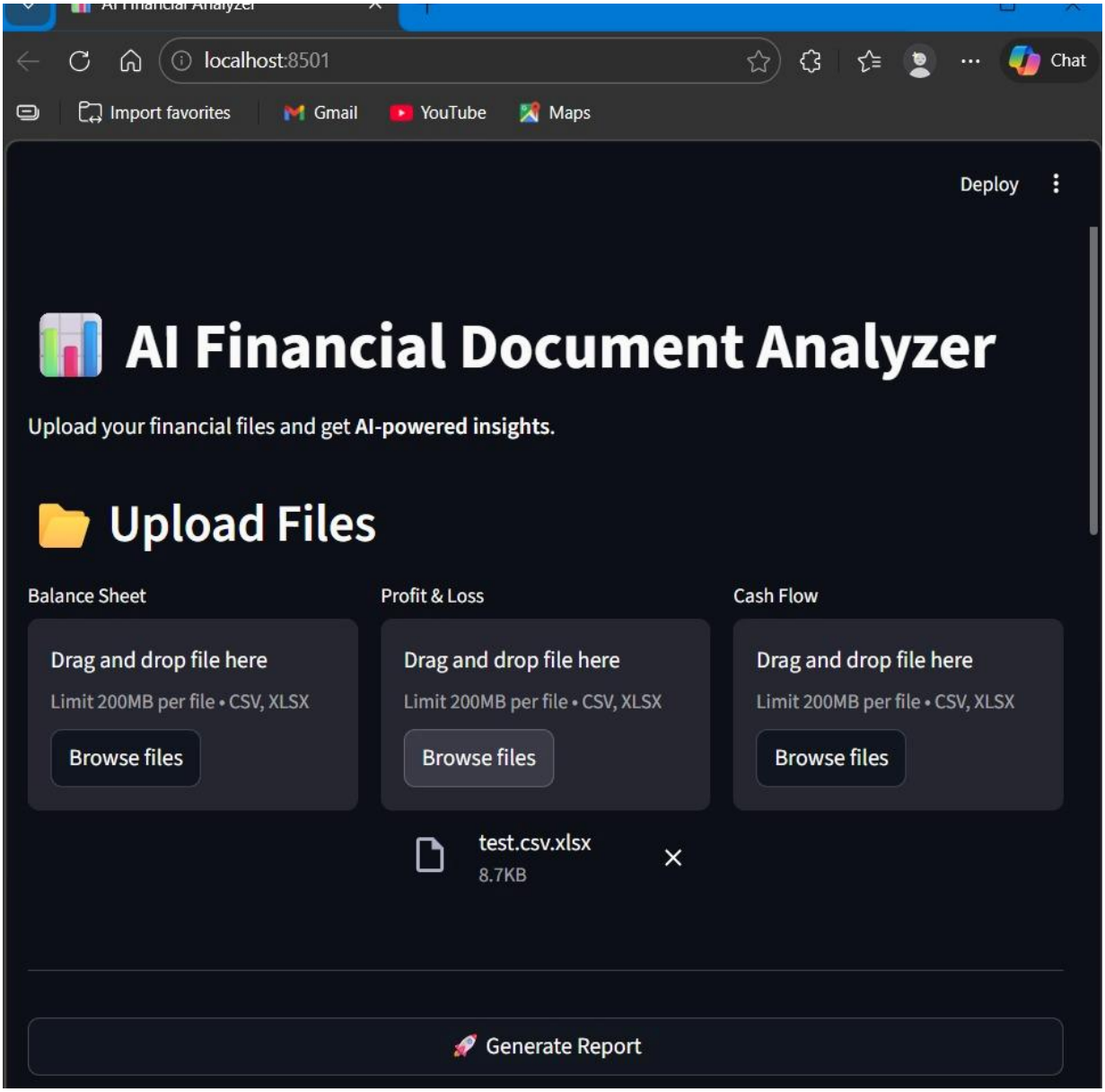
The Financial AI Analyzer system successfully achieved the following:

- Analyzed financial datasets efficiently.
- Identified spending patterns and financial trends.
- Detected abnormal or suspicious transactions.
- Generated structured financial summaries.
- Displayed insights through interactive visualizations.

The system helped users understand their financial behavior and make better financial decisions through data-driven insights.

7.1 Financial AI Analyzer User Interface

- **Figure 7.1.1:** Financial AI Analyzer interface allowing users to upload financial datasets for analysis.
- **Figure 7.1.2:** Dashboard navigation section for viewing financial insights and charts.
- **Figure 7.1.3:** Interactive financial analysis dashboard displaying summaries and visualizations.



7.2 Primary Module – Financial Data Analysis

- **Figure 7.2.1:** Financial dataset analysis producing structured statistical output.
- **Figure 7.2.2:** Visualization of spending distribution using charts and graphs.

Profit & Loss Data

	Year	Revenue	Expense	Profit
0	2022	50000	30000	20000
1	2023	70000	40000	30000
2	2024	90000	55000	35000

7.3 Visualization Module

- **Figure 7.3.1:** Financial dashboard showing trends and financial summaries for user interpretation.

A	B	C	D
Year	Revenue	Expense	Profit
2022	50000	30000	20000
2023	70000	40000	30000
2024	90000	55000	35000

8. ADVANTAGES AND DISADVANTAGES

8.1 Advantages

1. **Intelligent Financial Analysis**
   The system automatically analyzes financial datasets and generates meaningful insights, helping users understand complex financial data.
2. **Automated Data Visualization**
   Financial insights are displayed using charts and graphs, making data interpretation easier and more intuitive.
3. **Efficient Data Processing**
   The system can process large datasets quickly and provide results within seconds.
4. **User-Friendly Interface**
   The application provides a simple and interactive interface that allows users to upload datasets and view insights easily.
5. **Scalable Architecture**
   The system is designed to be scalable and can be extended with additional financial analytics features in the future.

8.2 Disadvantages

1. **Data Quality Dependency**
   The accuracy of insights depends on the quality and completeness of the uploaded financial data.
2. **Limited Dataset Formats**
   Currently the system supports limited file formats such as CSV and Excel.
3. **Internet Dependency (for deployment)**
   Cloud deployment may require a stable internet connection.
4. **Limited Predictive Capability**
   The current system focuses mainly on financial analysis rather than advanced predictive financial forecasting.

9. CONCLUSION

The Financial AI Analyzer project successfully demonstrates the practical application of artificial intelligence and data analytics in financial data analysis.

The system allows users to upload financial datasets and automatically generate structured insights such as spending patterns, transaction trends, anomaly detection, and financial summaries. By combining data processing techniques with visualization tools, the system converts complex financial data into easy-to-understand insights.

The application improves financial awareness and supports better financial decision-making by providing clear graphical representations and analytical reports.

Overall, the project highlights how AI-driven financial analytics can simplify financial data interpretation and enhance decision-support systems in personal and organizational finance.

10. FUTURE SCOPE

The Financial AI Analyzer system can be enhanced further with additional advanced features.

1. **Real-Time Financial Data Integration**
   Integration with banking APIs or financial services could allow real-time financial analysis.
2. **Predictive Financial Forecasting**
   Machine learning models could be added to predict future financial trends and spending behavior.
3. **Fraud Detection Module**
   Advanced anomaly detection algorithms could be implemented to identify fraudulent financial activities.
4. **Mobile Application Development**
   Developing a mobile application version would improve accessibility and usability.
5. **Multi-Language Support**
   Adding regional language support would make the application more accessible to a wider user base.
6. **Automated Financial Recommendations**
   The system could suggest personalized financial planning strategies based on user data.
7. **Advanced Visualization Dashboard**
   Future versions could include interactive dashboards with deeper financial analytics and reporting tools.

11. APPENDIX

11.1 Project Repository

- **GitHub Link(Team Lead) :** https://github.com/yallajosyula/financial-ai-analyzer
- **GitHub Link(Team Member) :** https://github.com/kaveri3tammineni/financial-ai-analyzer

11.2 Video Demonstration

- **Project Demo Video : https://drive.google.com/file/d/1zwQ2-NDh2XLvpw8Zx3M7MwOUvgD_x3CM/view?usp=drive_link**

11.3 Documentation References

- **Python Official Documentation: https://docs.python.org/3/**
- **Streamlit Documentation:** https://docs.streamlit.io/
- **Pandas Documentation:** https://pandas.pydata.org/docs/
- **Matplotlib Documentation:** https://matplotlib.org/stable/index.html

11.4 Code Overview

11.4.1 Requirements File

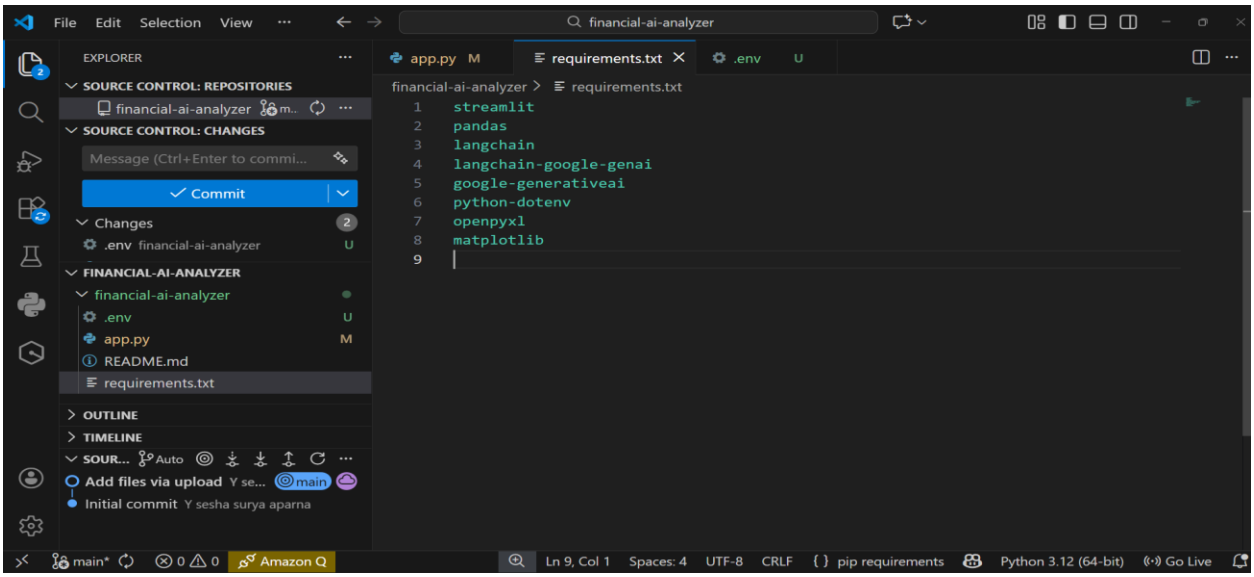
The following libraries are required to run the **Financial AI Analyzer** application. These dependencies are listed in the **requirements.txt** file.

Example libraries include:

- streamlit

- pandas
- numpy
- matplotlib
- python-dotenv

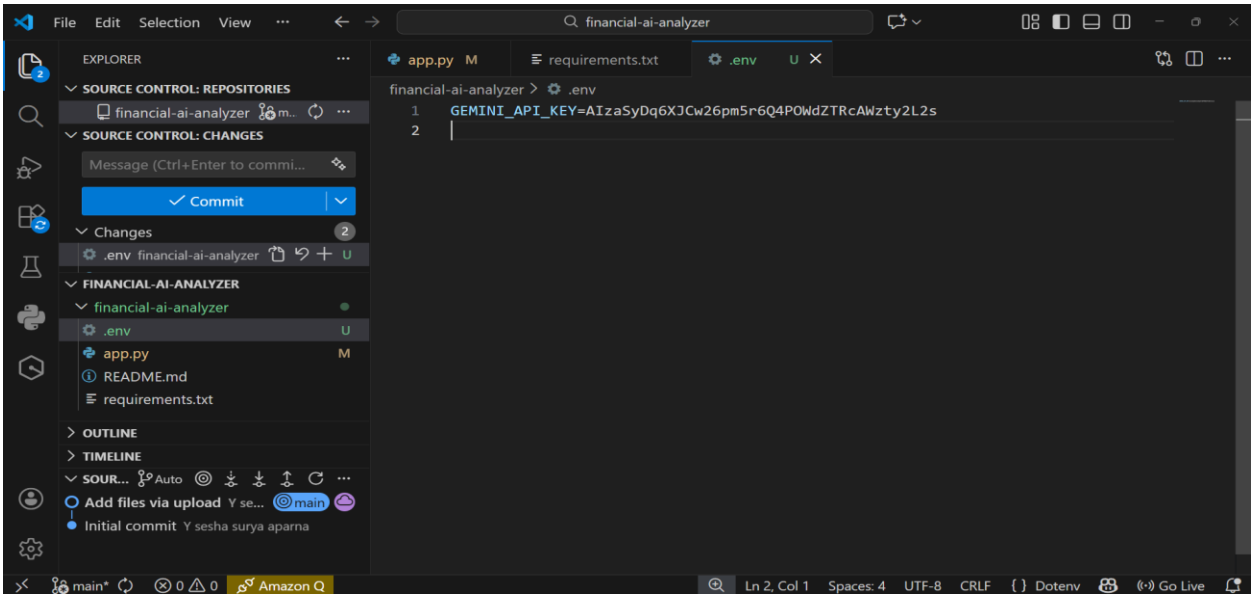
These libraries support data processing, visualization, and web interface development.



11.4.2 Environment Configuration

The application uses an environment configuration file **(.env)** to securely store configuration variables if required.

This approach prevents sensitive information or configuration parameters from being hardcoded directly into the source code and ensures secure development practices.

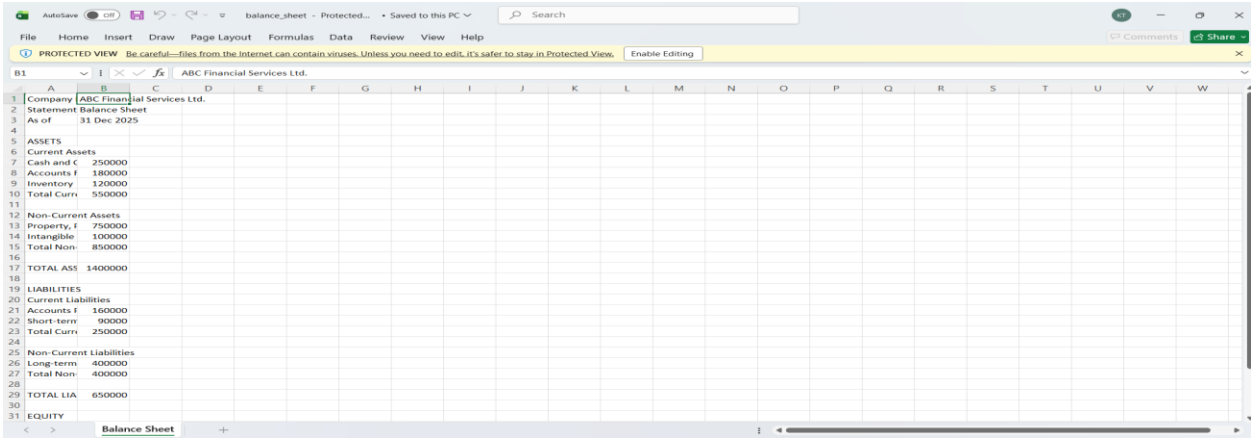


11.4.3 Core Code Snippets

1. Data Upload Initialization

This code initializes the dataset upload functionality. It allows users to upload financial datasets such as CSV or Excel files through the Streamlit interface.

The system reads the uploaded dataset and prepares it for further processing and analysis.



2. Data Loading and Processing

This module loads the uploaded dataset into a structured format using the **Pandas library**. It performs initial preprocessing tasks such as:

- Handling missing values

- Formatting columns
- Preparing data for analysis

This step ensures the dataset is ready for financial analysis.

```
financial-ai-analyzer > app.py > ...
import streamlit as st
import pandas as pd
import os
from dotenv import load_dotenv
from google import genai
```

3. Financial Data Analysis Logic

This component performs the core financial analysis. It processes the dataset to extract meaningful insights such as:

- Total income and expenses
- Spending patterns
- Category-wise expenditure
- Financial trends

The system calculates statistical summaries that help users understand their financial behavior.

```
financial-ai-analyzer > app.py > ...

def load_file(file):
    if file is None:
        return None

    try:
        if file.name.endswith(".csv"):
            df = pd.read_csv(file)

        elif file.name.endswith(".xlsx"):
            df = pd.read_excel(file)

        else:
            return None

        # Make Streamlit-safe
        df = df.fillna("")
        for col in df.columns:
            df[col] = df[col].astype(str)
```

4. Visualization and Chart Generation

This module generates graphical representations of financial data using **Matplotlib**. The charts help users visualize financial patterns and trends more clearly.

Examples include:

- Spending distribution charts
- Category-wise expense graphs
- Trend analysis charts

These visualizations make financial insights easier to interpret.

```
financial-ai-analyzer > app.py > visualize

# ================================
# VISUALIZATION
# ================================

def visualize(title, data):
    st.subheader(title)

    if data is not None:
        st.dataframe(data, width="stretch")

        numeric = data.apply(pd.to_numeric, erro

        if not numeric.dropna(axis=1, how="all")
            st.line_chart(numeric)
        else:
            st.info("No numeric data available")
    else:
        st.warning("No data uploaded")
```

5. User Interface Integration

This code manages the interaction between the user interface and the backend analysis modules.

The Streamlit interface enables users to:

- Upload datasets
- View financial summaries
- Explore charts and visualizations
- Analyze financial insights

This integration ensures smooth communication between the frontend interface and backend processing modules.

```
financial-ai-analyzer > app.py > ...
# UI
st.set_page_config(
    page_title="AI Financial Analyzer",
    page_icon="📊",
    layout="wide"
)

st.title("📊 AI Financial Document Analyzer")
st.markdown("Upload your financial files and get AI-powered insights.")

st.header("📂 Upload Files")

col1, col2, col3 = st.columns(3)

with col1:
    balance_file = st.file_uploader("Balance Sheet", ["csv", "xlsx"])

with col2:
    profit_file = st.file_uploader("Profit & Loss", ["csv", "xlsx"])

with col3:
    cash_file = st.file_uploader("Cash Flow", ["csv", "xlsx"])

st.markdown("---")
```

6. Financial Insights Display

This module formats the analyzed results and presents them in an organized and readable format.

The system displays:

- Financial summaries
- Statistical insights
- Visual charts and graphs

This helps users easily interpret the results and make informed financial decisions.

financial-ai-analyzer > app.py > ...

```python
            "Cash Flow"
        ])

        with tab1:
            st.write(balance_summary)

        with tab2:
            st.write(profit_summary)

        with tab3:
            st.write(cash_summary)

        st.header("📈 Visualizations")

        visualize("Balance Sheet Data", balance_data)
        visualize("Profit & Loss Data", profit_data)
        visualize("Cash Flow Data", cash_data)

st.markdown("---")
st.markdown(
    "<center>Developed by Appu using Streamlit & Gemini AI ❤️</center>",
    unsafe_allow_html=True
)
```